

TECHNOVA ELECTRONICS INC.

PRODUCTION DEVIATION
REPORT

PRODUCT	TechNova WorkStation 16 Laptop (FG-LP-WRK16)
PLANT	Austin Assembly Plant
DEVIATION PERIOD	August 1–31, 2025
REPORT ID	PDR-2025-08-001
PREPARED BY	Manufacturing Engineering
DATE	September 10, 2025

FIRST PASS YIELD: 90.2%
(TARGET 97%)

CLASSIFICATION: INTERNAL USE ONLY

Deviation Summary

The TechNova WorkStation 16 Laptop experienced the lowest first pass yield of all laptop models during August 2025, averaging **90.2%** against a target of 97%. This represents a significant deviation requiring root cause analysis and corrective action.

Production Overview (WorkStation 16)

Metric	Value
Total Production Orders	87
Total Planned Quantity	1,198 units
Good Quantity Produced	1,135 units
Scrap Quantity	5 units
Rework Quantity	58 units
Average Cycle Time	2.0 days
Schedule Variance	+0.3 days average delay
Production Type Split	18% Make-to-Order, 82% Batch Production

Comparison with Other Models

Product	First Pass Yield	Cycle Time
TechNova AirBook 13 Laptop	99.6%	1.9 days
TechNova EduBook 14 Laptop	95.9%	2.0 days
TechNova ProBook 15 Laptop	94.9%	2.0 days
TechNova ProBook 14 Laptop	91.1%	2.1 days
TechNova WorkStation 16 Laptop ** DEVIATION **	90.2%	2.0 days

The WorkStation 16 yield is 9.4 percentage points below the AirBook 13, indicating a model-specific assembly issue rather than a systemic plant problem.

Root Cause Analysis

Primary Root Cause: Motherboard-to-chassis alignment tolerance

The WorkStation 16 uses a larger 16-inch chassis (Aluminum Chassis 16in) with tighter thermal tolerances than other models. Investigation revealed that 62% of rework cases involved motherboard mounting misalignment causing thermal sensor false positives during final test.

Contributing Factors

1. Component Factor

Aluminum Chassis 16in from CasingCraft Manufacturing showed dimensional variance of $\pm 0.3\text{mm}$ on mounting holes (spec: $\pm 0.1\text{mm}$). CasingCraft has 32 total rejected units during this period with a 3.3% rejection rate.

2. Assembly Process Factor

The WorkStation 16 assembly jig was last calibrated in May 2025. Recommended calibration interval is 60 days. Jig deflection of 0.15mm was measured, contributing to alignment issues.

3. Testing Factor

Final test thermal threshold was set at 72°C, which is appropriate for standard models but too aggressive for the WorkStation 16's higher TDP processor. 18 final test failures were recorded for the AirBook 13 and 18 for the ProBook 15 during the same period.

Fishbone Diagram Summary: Machine — assembly jig out of calibration; Material — chassis dimensional variance from CasingCraft; Method — thermal test threshold not model-specific; Manpower — operator training gap on WorkStation 16 specifics.

Financial Impact

Direct Costs

Item	Amount / Detail
Rework labor (58 units x 1.5 hrs x \$35/hr)	\$3,045
Scrap write-off (5 units x \$1,200 standard cost)	\$6,000
Additional machine hours (58 units x 0.5 hrs)	29 hours
Material waste from scrapped units	\$2,800

Indirect Costs

Item	Amount / Detail
Schedule delay impact on 12 Make-to-Order customer orders	\$8,400 (est. customer credit notes and expedited shipping)
Production capacity lost to rework	87 machine hours (could have produced 29 additional units)
Quality team investigation time	120 person-hours

Total Estimated Cost of Deviation: \$20,245

Inventory Impact

The WorkStation 16 current inventory at Austin warehouse is 29 units with a status of “Low.” The book value is \$46,800. With only 2 units in QC hold, the available-to-promise quantity is 29 units. Combined with the yield issues, there is a risk of stockout within 2 weeks if demand continues at current levels.

Corrective Actions

Immediate (Implemented)

- ✓ Recalibrated WorkStation 16 assembly jig on September 2, 2025
- ✓ Adjusted thermal test threshold to 78°C for WorkStation 16 model

✓ Added incoming inspection step for chassis dimensional check

Short-Term (In Progress)

- CasingCraft Manufacturing corrective action request issued — response due September 25, 2025; requirement: tighten mounting hole tolerance to ±0.15mm
- Create model-specific assembly work instructions for WRK16
- Schedule operator refresher training (12 assemblers)

Long-Term (Planned)

- Implement automated optical alignment verification station — capital request \$85,000; expected installation Q1 2026; projected yield improvement +4% first pass yield
- Establish calibration tracking system with automated alerts
- Evaluate alternative chassis supplier for dual-sourcing

Effectiveness Verification: Target first pass yield ≥ 95% by November 2025. Monitoring: weekly yield tracking by model in Manufacturing dashboard. Escalation: if yield does not reach 93% by October 15, escalate to VP Ops.

Approval

Manufacturing Manager: _____ Date: _____

Quality Manager: _____ Date: _____

Plant Director: _____ Date: _____